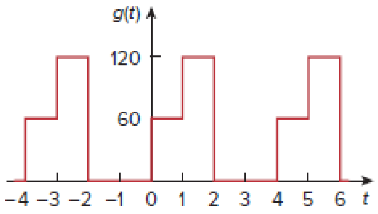
Chapter 17, Problem 3P

Problem

Give the Fourier coefficients *a*0, *an*, and *bn* of the waveform in Fig. 17.47. Plot the amplitude and phase spectra.

Figure 17.47:



Step-by-step solution

Step 1 of 12

Refer to waveform in Figure 17.47 in the textbook.

The signal $g(t)$ is,

$$g(t)=\begin{cases}60 & 0<t<1\\ 120 & 1<t<2\\ 0 & 2<t<4\end{cases}$$

From the waveform in Figure 17.47, observe that the signal is repeated for every 4 s. Hence the time period is $T=4\text{ s}$.

Calculate the fundamental analog frequency.

$$\begin{aligned}\omega_0&=\frac{2\pi}{T}\\&=\frac{2\pi}{4}\\&=\frac{\pi}{2}\end{aligned}$$

Step 2 of 12

Calculate the Fourier coefficient value a_0.

$$\begin{aligned}a_0&=\frac{1}{T}\int_0^T g(t)\,dt\\&=\frac{1}{4}\int_0^4 f(t)\,dt\\&=(0.25)\left(\int_0^1(60)\,dt+\int_1^2(120)\,dt+\int_2^4(0)\,dt\right)\\&=(0.25)\left[(60)(1-0)+(120)(2-1)\right]\end{aligned}$$

$$\begin{aligned}&=(0.25)[60+120]\\&=45\end{aligned}$$

Step 3 of 12

Calculate the Fourier coefficient value a_n.

$$a_n = \frac{2}{T}\int_0^T g(t)\cos n\omega_0 t dt$$
$$= \frac{2}{4}\int_0^4 g(t)\cos n\left(\frac{\pi}{2}\right)t dt$$
$$= \frac{1}{2}\left(\int_0^1 60\cos n\left(\frac{\pi}{2}\right)t dt + \int_1^2\left(120\cos n\left(\frac{\pi}{2}\right)t\right)dt + \int_2^4\left(0\cos n\left(\frac{\pi}{2}\right)t\right)dt\right)$$
$$= (30)\left(\frac{\sin n\left(\frac{\pi}{2}\right)t}{n\left(\frac{\pi}{2}\right)}\right)\Bigg|_{t=0}^{1} + (60)\left(\frac{\sin n\left(\frac{\pi}{2}\right)t}{n\left(\frac{\pi}{2}\right)}\right)\Bigg|_{t=1}^{2}$$
$$= \frac{30}{n\left(\frac{\pi}{2}\right)}\left(\sin\frac{n\pi}{2} - \sin\frac{n}{2}(0)\right) + \frac{60}{n\left(\frac{\pi}{2}\right)}\left(\sin\frac{2n\pi}{2} - \sin\frac{n\pi}{2}\right)$$
$$= \left(\frac{60}{n\pi}\right)\sin\frac{n\pi}{2} + \frac{120}{n\pi}\left(-\sin\frac{n\pi}{2}\right)$$
$$= -\left(\frac{60}{n\pi}\right)\sin\left(\frac{n\pi}{2}\right)$$

The value of the Fourier coefficient a_n for the signal $g(t)$ is,

$$a_n = \begin{cases} \frac{-60}{n\pi}(-1)^{\frac{n+1}{2}} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

Step 4 of 12

Calculate the Fourier coefficient value b_n.

$$b_n = \frac{2}{T}\int_0^T g(t)\sin n\omega_0 t dt$$
$$= \left(\frac{2}{4}\right)\int_0^4 g(t)\sin n\left(\frac{\pi}{2}\right)t dt$$
$$= \frac{1}{2}\left(\int_0^1 60\sin n\left(\frac{\pi}{2}\right)t dt + \int_1^2\left(120\sin n\left(\frac{\pi}{2}\right)t\right)dt + \int_2^4\left(0\sin n\left(\frac{\pi}{2}\right)t\right)dt\right)$$
$$= (0.5)\left((60)\left(\frac{-\cos n\left(\frac{\pi}{2}\right)t}{n\left(\frac{\pi}{2}\right)}\right)\Bigg|_{t=0}^{1} + (120)\left(\frac{-\cos n\left(\frac{\pi}{2}\right)t}{n\left(\frac{\pi}{2}\right)}\right)\Bigg|_{t=1}^{2}\right)$$
$$= \frac{0.5}{n\left(\frac{\pi}{2}\right)}\left(-60\left(\cos n\left(\frac{\pi}{2}\right) - \cos n\left(\frac{\pi}{2}\right)(0)\right) - 120\left(\cos 2n\left(\frac{\pi}{2}\right) - \cos n\left(\frac{\pi}{2}\right)\right)\right)$$
$$= \frac{1}{n\pi}\left(-60\left(\cos n\left(\frac{\pi}{2}\right) - 1\right) - 120\left(\cos n\pi - \cos n\left(\frac{\pi}{2}\right)\right)\right)$$
$$= \frac{60}{n\pi}\left(1 + \cos n\left(\frac{\pi}{2}\right) - 2\cos n\pi\right)$$

Step 5 of 12

The Fourier series is,

$$g(t)=a_0+\sum_{n=1}^{\infty}\left(a_n\cos n\omega_0 t+b_n\sin n\omega_0 t\right)$$
$$=45+\sum_{n=1}^{\infty}\left(a_n\cos n\omega_0 t+b_n\sin n\omega_0 t\right)$$
$$=45+\frac{60}{\pi}\sum_{n=1}^{\infty}\left\{\frac{1}{n}\left[1+\cos\left(\frac{n\pi}{2}\right)-2\cos(n\pi)\right]\sin\left(\frac{n\pi}{2}t\right)-\frac{1}{n}\sin\left(\frac{n\pi}{2}\right)\cos\left(\frac{n\pi}{2}t\right)\right\}$$

The amplitude of n-th harmonic is,

$$A_n=\sqrt{a_n^2+b_n^2},\quad n \text{ is odd}$$
$$=\sqrt{\left[-\frac{60}{n\pi}\sin\left(\frac{n\pi}{2}\right)\right]^2+\left\{\frac{60}{n\pi}\left[1+\cos\left(\frac{n\pi}{2}\right)-2\cos(n\pi)\right]\right\}^2}$$
$$=\frac{60}{n\pi}\sqrt{\left[\sin\left(\frac{n\pi}{2}\right)\right]^2+\left[1+\cos\left(\frac{n\pi}{2}\right)-2\cos(n\pi)\right]^2}$$

Step 6 of 12

Calculate the first harmonic.

$$A_1=\frac{60}{\pi}\sqrt{\left[\sin\left(\frac{\pi}{2}\right)\right]^2+\left[1+\cos\left(\frac{\pi}{2}\right)-2\cos(\pi)\right]^2}$$
$$=60.3951$$

Calculate the second harmonic.

$$A_2=\frac{60}{2\pi}\sqrt{\left[\sin\left(\frac{2\pi}{2}\right)\right]^2+\left[1+\cos\left(\frac{2\pi}{2}\right)-2\cos(2\pi)\right]^2}$$
$$=19.0986$$

Calculate the third harmonic.

$$A_3=\frac{60}{3\pi}\sqrt{\left[\sin\left(\frac{3\pi}{2}\right)\right]^2+\left[1+\cos\left(\frac{3\pi}{2}\right)-2\cos(3\pi)\right]^2}$$
$$=20.1317$$

Step 7 of 12

Calculate the fourth harmonic.

$$A_4=\frac{60}{4\pi}\sqrt{\left[\sin\left(\frac{4\pi}{2}\right)\right]^2+\left[1+\cos\left(\frac{4\pi}{2}\right)-2\cos(4\pi)\right]^2}$$
$$=0$$

Calculate the fifth harmonic.

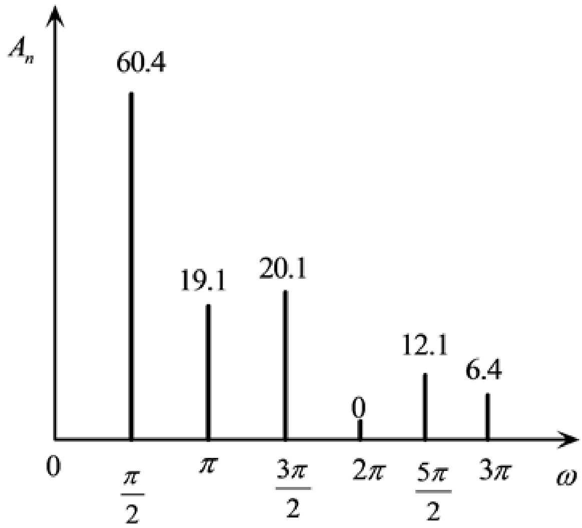
$$A_5=\frac{60}{5\pi}\sqrt{\left[\sin\left(\frac{5\pi}{2}\right)\right]^2+\left[1+\cos\left(\frac{5\pi}{2}\right)-2\cos(5\pi)\right]^2}$$
$$=12.079$$

Calculate the sixth harmonic.

$$A_6=\frac{60}{6\pi}\sqrt{\left[\sin\left(\frac{6\pi}{2}\right)\right]^2+\left[1+\cos\left(\frac{6\pi}{2}\right)-2\cos(6\pi)\right]^2}$$
$$=6.3662$$

Step 8 of 12

The amplitude spectrum is shown in Figure 1.



Therefore, the amplitude spectrum is shown in Figure 1.

Step 9 of 12

The Phase ϕ_n expression is

$$\begin{aligned}\phi_n &= -\tan^{-1}\left(\frac{b_n}{a_n}\right)\\ &= -\tan^{-1}\left(\frac{\frac{60}{n\pi}\left[1+\cos\left(\frac{n\pi}{2}\right)-2\cos(n\pi)\right]}{-\frac{60}{n\pi}\sin\left(\frac{n\pi}{2}\right)}\right)\\ &= \tan^{-1}\left(\frac{1+\cos\left(\frac{n\pi}{2}\right)-2\cos(n\pi)}{\sin\left(\frac{n\pi}{2}\right)}\right)\end{aligned}$$

Calculate phase ϕ_1.

$$\begin{aligned}\phi_1 &= \tan^{-1}\left(\frac{1+\cos\left(\frac{\pi}{2}\right)-2\cos(\pi)}{\sin\left(\frac{\pi}{2}\right)}\right)\\ &= \tan^{-1}(3)\\ &= 71.5651^\circ\end{aligned}$$

Step 10 of 12

Calculate phase ϕ_2.

$$\phi_2 = \tan^{-1}\left(\frac{1+\cos\left(\frac{2\pi}{2}\right)-2\cos(2\pi)}{\sin\left(\frac{2\pi}{2}\right)}\right)$$
$$= -90°$$

Calculate phase ϕ_3.

$$\phi_3 = \tan^{-1}\left(\frac{1+\cos\left(\frac{3\pi}{2}\right)-2\cos(3\pi)}{\sin\left(\frac{3\pi}{2}\right)}\right)$$
$$= \tan^{-1}(-3)$$
$$= -71.5651°$$

Calculate phase ϕ_4.

$$\phi_4 = \tan^{-1}\left(\frac{1+\cos\left(\frac{4\pi}{2}\right)-2\cos(4\pi)}{\sin\left(\frac{4\pi}{2}\right)}\right)$$
$$= \text{undefined}$$

Step 11 of 12

Calculate phase ϕ_5.

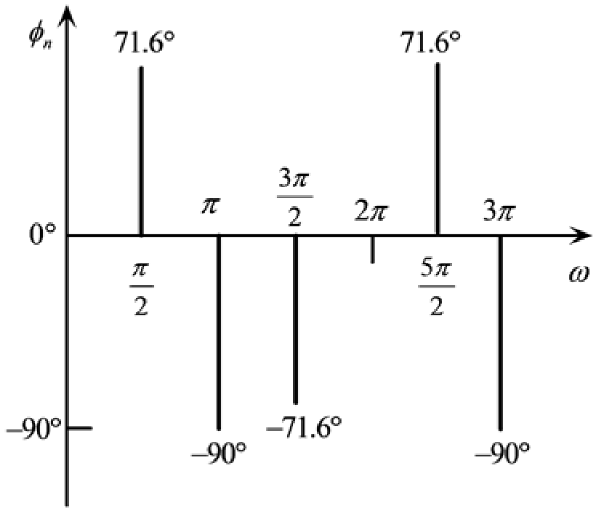
$$\phi_5 = \tan^{-1}\left(\frac{1+\cos\left(\frac{5\pi}{2}\right)-2\cos(5\pi)}{\sin\left(\frac{5\pi}{2}\right)}\right)$$
$$= \tan^{-1}(3)$$
$$= 71.5651°$$

Calculate phase ϕ_6.

$$\phi_6 = \tan^{-1}\left(\frac{1+\cos\left(\frac{6\pi}{2}\right)-2\cos(6\pi)}{\sin\left(\frac{6\pi}{2}\right)}\right)$$
$$= \tan^{-1}(-\infty)$$
$$= -90°$$

Step 12 of 12

The phase spectrum is shown in Figure 2.



Therefore, the phase spectrum is shown in Figure 2.